

Lecture 7: Behavioural Game Theory

Level-k Thinking, the Beauty Contest, and Why Nash Equilibrium Often Fails

Prof. Muhebullah Karimzada

School of Economics, University of Northern British Columbia

ECON 230 · Introduction to Behavioural Economics · Winter 2027

The Keynesian Beauty Contest

Guess a number between 0 and 100.

The winner is the person whose guess is **closest to $\frac{2}{3}$ of the average guess** across all players.

Write your answer on a piece of paper. Do not discuss with neighbours. Hand it to the front.

Why “Beauty Contest”?

Keynes (1936) described stock market investing as a beauty contest where you don't pick who *you* find most beautiful, but who you think *others* will find most beautiful. The same logic applies here: don't guess what *you* think is right — guess what you think others will guess (times $\frac{2}{3}$).

1. **Nash equilibrium** and what it requires
2. Level-k thinking — strategic reasoning in steps
3. Beauty Contest debrief — where does your class cluster?
4. The Centipede Game — backward induction failures
5. The Prisoner's Dilemma — cooperation vs defection
6. Repeated games and cooperation — Folk Theorem, Tit-for-Tat
7. Quantal Response Equilibrium — noisy best responses
8. Applications: auctions, oligopoly, negotiations

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Nash Equilibrium — What It Actually Requires

Definition

A strategy profile $(s_1^*, \dots, s_n^*)$ is a Nash Equilibrium if no player can unilaterally improve their payoff by deviating.

What Nash Equilibrium Requires

1. Everyone is **rational**.
2. Everyone knows that everyone is rational.
3. Everyone knows that everyone knows that everyone is rational.
4. *Ad infinitum*. This is **Common Knowledge of Rationality (CKR)**.

This is an extremely strong assumption.

The Empirical Failure

CKR fails whenever:

- Players have limited depths of reasoning (most people: 1–2 steps)
- Players have social preferences (Lecture 6)
- Players have bounded rationality (Lecture 1)
- The game is unfamiliar

Nash equilibrium is a long-run benchmark, not a description of initial play

The Prisoner's Dilemma

The Game

	Cooperate	Defect
C	(3, 3)	(0, 5)
D	(5, 0)	(1, 1) Nash

Analysis

- D **dominates** C for both players.
- Nash: (D,D) — both get 1.
- Pareto optimum: (C,C) — both get 3.
- Cooperation is *not* a Nash equilibrium in a one-shot game.

Real-World PDs

- Airline price wars; OPEC cartel
- Nuclear arms races; advertising wars
- Antibiotic resistance; climate change

[GAME] Prisoner's Dilemma in Class

Rules

1. Pair with your neighbour. You will play 5 rounds with the **same** partner.
2. Each round: simultaneously write **C** (cooperate) or **D** (defect).
3. Payoffs: $(C,C) = (3,3)$. $(C,D) = (0,5)$. $(D,C) = (5,0)$. $(D,D) = (1,1)$.
4. Reveal after each round. Keep a running score.

Debrief Questions

- How many pairs ended up at (C,C) ? (D,D) ?
- Did strategy evolve over the 5 rounds?
- Did you use Tit-for-Tat? Did it work?
- How did it feel to be defected on after cooperating?

The Level-k Model

Stahl & Wilson (1994); Nagel (1995): Instead of CKR (infinite depth), players have types by reasoning depth:

Level Types:

- **L0:** Uniform random or non-strategic.
- **L1:** Best respond to L0 belief.
- **L2:** Best respond to L1 belief.
- **Nash:** $k \rightarrow \infty$.

Beauty Contest:

- **L0** plays **50**. **L1** plays **33**. **L2** plays **22**.
- **L3:** 15. **L4:** 10. **Nash:** 0.

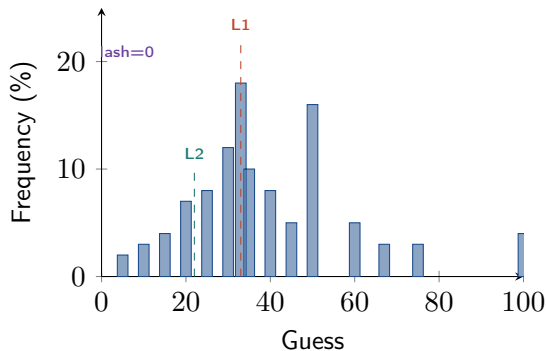
What Empirical Data Shows

Spikes at 33 (L1) and 22 (L2). Few at 0. **Most people are L1–L2.** FT competition (1,500 readers): winning guess = 13 ($\approx$ L3). Camerer et al. (2004): average $k \approx 1.5$.

Beauty Contest Debrief — Your Class Results

Compute: (1) class average; (2) $\frac{2}{3} \times$ average; (3) find the winner; (4) plot distribution.

Typical Beauty Contest Distribution



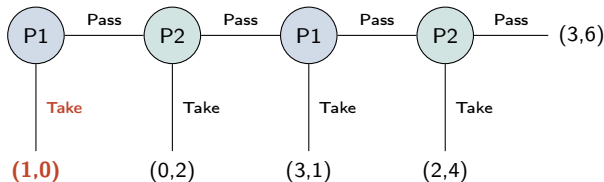
Nagel (1995) Data

- Caltech: mean ≈ 34 (L1). Grad students: ≈ 25 (L2).
- FT competition (1,500 readers): 13.
- Science readers (Germany): ≈ 22 .

The Centipede Game

Rosenthal (1981)

Two players alternate moves. At each node: **Take** (game ends, payoffs as below) or **Pass** (game continues, pot doubles). Standard backward induction: Player 1 should Take at node 1!



The Prediction vs Reality

Nash: Player 1 takes immediately $\Rightarrow$ both get $(1, 0)$.

Reality (McKelvey & Palfrey, 1992): Games almost *never* end at node 1. Most go to node 3 or 4. Why? Social preferences, bounded rationality, level-k reasoning.

Cooperation in Repeated Games

The Folk Theorem

In an infinitely repeated game, any feasible, individually rational payoff can be sustained as a Nash Equilibrium with sufficiently patient players (δ close enough to 1). This includes the cooperative (C,C) outcome!

Tit-for-Tat (Axelrod, 1984)

Robert Axelrod ran computer tournaments where strategies competed in repeated PD.

Winner (twice): Tit-for-Tat.

- **Nice:** Cooperate on round 1.
- **Retaliatory:** Copy opponent's last move.
- **Forgiving:** Return to C if opponent returns to C.
- **Clear:** Simple to understand.

Real-World Cooperation

- **OPEC:** Repeated interaction + monitoring sustains cartel.
- **Canada-US trade:** 150 years of largely cooperative relationship.
- **Indigenous ROSCAs:** Rotating savings groups in immigrant communities.
- **Biology:** Bacteria cooperate in biofilms despite defection incentives.

McKelvey & Palfrey (1995)

Instead of fully rational choice (always pick best response), players choose better options **more often but not always**:

$$\Pr(\text{action } a) \propto \exp(\lambda \cdot EU(a))$$

- $\lambda = 0$: Uniform random (fully irrational)
- $\lambda \rightarrow \infty$: Nash equilibrium (fully rational best response)
- Intermediate λ : “noisy best-responder”

Why QRE Matters

QRE fits experimental data **far better** than Nash in centipede games, coordination games, entry games, and ultimatum games. Nash is the limit of QRE as rationality $\lambda \rightarrow \infty$ — QRE nests Nash as a special case.

In a **common value auction**, the winner has the most optimistic signal — so the true value is likely *below* their estimate. The winner is **cursed**.

Rational Bidding

Shade bid *below* your estimate to correct for the curse. Naive bidding: overbid by 15–30% (Kagel & Levin, 1986).

Real Examples

- **FCC auctions:** Milgrom & Wilson (Nobel 2020).
- **Canadian oil sands:** Overbidding in 1970s.
- **Corporate M&A:** Acquirers typically overpay.
- **IPOs:** Retail get what professionals reject.

Discussion Questions

- Q1. “If you were designing a government procurement auction, how would you account for winner’s curse? What rules would minimise overbidding?”
- Q2. “Can you think of a real-world ‘beauty contest’ situation where being wrong in the *same direction as everyone else* is safer than being right alone?” (Hint: fund manager career concerns; housing market)
- Q3. “Canada’s telecom market (Bell, Rogers, Telus): is this repeated-game tacit collusion, competitive equilibrium, or something else? What evidence would distinguish these?”

Key Takeaways — Lecture 7

- ✓ **Nash requires CKR** — common knowledge of rationality to infinite depth. Unrealistic.
- ✓ **Level-k model**: L_0 = random, L_1 = best respond to L_0 , ...; most people are L_1 – L_2 .
- ✓ **Beauty contest**: Nash = 0, typical winning guess = 13–33 depending on audience.
- ✓ **Centipede**: backward induction fails; cooperation near start of game.
- ✓ **Repeated games**: cooperation supported by Folk Theorem, Tit-for-Tat, social norms.
- ✓ **QRE**: noisy best-response; nests Nash; fits data better in lab.
- ✓ **Applications**: auctions (winner's curse), oligopoly (tacit collusion), M&A (overbidding).

Next: Lecture 8: Behavioural Finance — How individual biases aggregate into market bubbles, crashes, and persistent anomalies.